

NYC 311 Service Requests Data Quality Rulebook

Rule Classification

Critical - Violations impact data integrity, record identification, or business process reporting.

High - Violations impact analytics, location reporting, and operational decision-making.

Medium - Violations impact consistency and standardization.

Low - Violations impact presentation and reporting quality.

DQ-001 Unique Service Request Identifier

Severity: Critical

Business Rule: Every service request must possess a unique unique_key.

Validation Logic: unique_key must be unique.

Failure Condition: Duplicate unique_key detected.

Remediation: Retain original record and investigate source-system duplication.

DQ-002 Created Date Mandatory

Severity: Critical

Business Rule: Every service request must have a valid created_date.

Validation Logic: created_date IS NOT NULL

Failure Condition: Missing creation timestamp.

Remediation: Reject record from analytical layer and log for investigation.

DQ-003 Valid Date Format

Severity: Critical

Business Rule: created_date, closed_date and due_date must be valid datetime values.

Validation Logic: Dates must successfully convert to DATETIME.

Failure Condition: Invalid datetime value.

Remediation: Attempt format standardization. Otherwise route to exception table.

DQ-004 Closed Date Must Not Precede Created Date

Severity: Critical

Business Rule: A service request cannot be closed before it is created.

Validation Logic: `closed_date >= created_date`

Failure Condition: Negative resolution duration.

Remediation: Flag record for manual review.

DQ-005 Due Date Must Not Precede Created Date

Severity: High

Business Rule: Resolution due date must occur on or after creation date.

Validation Logic: `due_date >= created_date`

Failure Condition: Impossible SLA timeline.

Remediation: Correct using source-system reference when available.

DQ-006 Future Date Validation

Severity: High

Business Rule: `created_date` and `closed_date` cannot occur in the future.

Validation Logic:

`created_date <= CURRENT_DATE`

`closed_date <= CURRENT_DATE`

Failure Condition: Future-dated service requests.

Remediation: Flag for investigation.

DQ-007 Closed Status Consistency

Severity: Critical

Business Rule: Closed requests must contain a closed_date.

Validation Logic:

```
status='Closed'  
AND closed_date IS NOT NULL
```

Failure Condition: Closed request with missing closure date.

Remediation: Investigate source-system update failure.

DQ-008 Open Status Consistency

Severity: High

Business Rule: Open requests should not possess a closed_date.

Validation Logic:

```
status<>'Closed'  
AND closed_date IS NULL
```

Failure Condition: Request marked open but already closed.

Remediation: Review workflow status.

DQ-009 Latitude Validity

Severity: High

Business Rule: Latitude values must fall within valid geographic bounds.

Validation Logic: latitude BETWEEN -90 AND 90

Failure Condition: Invalid coordinate.

Remediation: Nullify invalid value and flag record.

DQ-010 Longitude Validity

Severity: High

Business Rule: Longitude values must fall within valid geographic bounds.

Validation Logic: longitude BETWEEN -180 AND 180

Failure Condition: Invalid coordinate.

Remediation: Nullify invalid value and flag record.

DQ-011 Coordinate Pair Completeness

Severity: High

Business Rule: Latitude and longitude must exist together.

Validation Logic:

(latitude IS NULL AND longitude IS NULL)

OR

(latitude IS NOT NULL AND longitude IS NOT NULL)

Failure Condition: Only one coordinate available.

Remediation: Move to exception table.

DQ-012 Borough Standardization

Severity: Medium

Business Rule: Borough values must conform to approved borough master list.

Approved Values:

BRONX

BROOKLYN

MANHATTAN

QUEENS

STATEN ISLAND

Failure Condition: Any non-approved variation.

Examples Identified:

Brooklyn

BROOKLYN

Brooklyn NY

Remediation: Apply standardization mapping.

DQ-013 City Standardization

Severity: Medium

Business Rule: City names must follow approved reference values.

Examples Identified:

NYC
NEW YORK
Ny

Target Value: NEW YORK

Remediation: Reference mapping table.

DQ-014 Complaint Type Standardization

Severity: Medium

Business Rule: Complaint categories must use standardized naming.

Examples Identified:

Heat/Hot Water
HEAT/HOT WATER

Remediation: Standardized lookup table.

DQ-015 Blank String Validation

Severity: Medium

Business Rule: Blank strings are treated as missing values.

Validation Logic: TRIM(column) <> ''

Applies To:

agency
city
borough
complaint_type
status

Remediation: Convert blank strings to NULL.

DQ-016 Leading and Trailing Whitespace

Severity: Low

Business Rule: Text values must not contain leading or trailing whitespace.

Validation Logic: column = TRIM(column)

Remediation: Apply TRIM function.

DQ-017 Complaint Type Mandatory

Severity: High

Business Rule: Every request must contain a complaint_type.

Validation Logic: complaint_type IS NOT NULL

Failure Condition: Missing complaint category.

Remediation: Assign Unknown or investigate source record.

DQ-018 Agency Mandatory

Severity: High

Business Rule: Every request must contain an agency.

Validation Logic: agency IS NOT NULL

Failure Condition: Missing responsible agency.

Remediation: Investigate source assignment process.

DQ-019 ZIP Code Format Validation

Severity: Medium

Business Rule: Incident ZIP code must contain five numeric digits.

Validation Logic: ZIP matches approved ZIP pattern.

Failure Condition: Invalid ZIP code.

Remediation: Standardize or flag record.

DQ-020 Geographic Consistency

Severity: High

Business Rule: City, Borough, ZIP Code and Coordinates should represent a valid geographic combination.

Validation Logic: Cross-reference against approved geographic lookup table.

Failure Condition: Location attributes conflict.

Remediation: Review and correct using authoritative location source.